

Name: _____ Date: _____

Collaborators: _____ Lab Section: _____

PRELAB EXERCISES (6 points)*This prelab must be completed and handed in to the lab TA at the start of the lab.***Question 1****2 points**

You are trying to see if there is a pattern to when spam arrives in your mailbox, so you count the number of spam emails received over the last month and bin them by time of day. Out of 1595 spam emails, 784 were received during the day, between 6 AM and 6 PM. Treating spam emails as Poisson distributed, do your data support the idea that spammers prefer one part of the day over the other? *Hint: add error bars to the data and see if there is overlap.*

Question 2**2 points**

You investigate further and subdivide the day into four equal time intervals. What conclusion do you draw from your data now, and why?

Time of day	No. of spam emails
midnight to 6:00 AM	383
6:01 AM to noon	401
noon to 6:00 PM	374
6:01 PM to midnight	447

Question 3**2 points**

You are a quality engineer for a paper company and must ensure that the area of A4 sheets being produced has less than 0.1% error. You analyze a sample of A4 paper and find the width and length to be, respectively, 210 ± 0.2 mm and 297 ± 0.3 mm. Does the company have a problem? Explain.

